

Juan Esteban Bustos Chilatra

Full-Stack Developer · IoT & Robotics · Applied AI
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PROFESSIONAL SUMMARY

Full-Stack Developer with 15+ projects in production, 1st place at a regional robotics competition and 3 papers under submission to IEEE — all built in the first year of university. Specialized in TypeScript · React · Node.js · PHP · Go · Python with AI integration (LLMs, agents, chatbots) and hardware (Arduino, ESP32). I ship production code, not demos.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, PHP, Python, Go, SQL, C/C++ (Arduino), Java, Lua
Frontend: React, Tailwind CSS, Bootstrap, Three.js, GSAP, Axios, SweetAlert, MediaPipe, Web Speech API
Backend: Node.js, PHP MVC, REST APIs, Composer
Databases: PostgreSQL, MySQL, Firebase, Supabase, SQLite
Mobile: Capacitor, Java (Android Studio)
Hardware / IoT: Arduino Uno, ESP32, Driver L298N, RFID, RTC, Bluetooth, PWM, Módulo SD
AI & Tools: Ollama, Claude Code, AI Chatbot, Hermes Agent, Open Claw, VS Code, Git, Linux, Windows
Security & Auth: OAuth (Google Login), User roles, Session validation, OWASP best practices

ACHIEVEMENTS & RECOGNITION

1st Place — 2nd Regional Robotics Competition	2025
3rd Place — Regional University Hackathon <i>Competed in my first semester against upper-year teams.</i>	2025
Speaker — 2nd Youth & Artificial Intelligence Conference	2025
Presenter — InnovaSoft Fair, Universidad Surcolombiana <i>Presented VIANova, BioSmart and EcoGuardian to the full university community.</i>	2026
Author — 3 papers under submission to IEEE <i>VIANova, BioSmart (GopherMind AI) and EcoGuardian.</i>	2026

PROJECTS

VIANova — Web Platform <i>TypeScript · React · AI Chatbot · Tailwind CSS · Production</i> - Designed and built a web platform for smart urban mobility optimization, deployed to production. - Integrated a specialized AI conversational assistant for real-time user guidance within the transport ecosystem. - Leading the writing of the scientific research paper derived from this project for IEEE submission.	2026 – Present
VIANova — Mobile App <i>Capacitor · TypeScript · React · Mobile UX</i>	2026 – Present

- Built and packaged the cross-platform native mobile version with Capacitor, adapting critical mobility features for mobile devices.
- Presented the technical features, performance and social impact of the solution at the InnovaSoft Fair, Universidad Surcolombiana.

EcoGuardian — Web Platform

2026 – Present

TypeScript · React · AI Chatbot · Environmental Metrics · Production

- Built a comprehensive platform for environmental monitoring and active ecological management, deployed to production.
- Implemented an AI Chatbot focused on resolving environmental queries and processing ecological alert reports.
- Leading the research paper writing for IEEE submission.

EcoGuardian — Mobile App

2026 – Present

Capacitor · TypeScript · Mobile UI · Interconnectivity

- Designed and deployed the native mobile app with Capacitor, enabling ecological incident reporting and critical metrics visualization from any device.

BioSmart (GopherMind AI)

2026 – Present

Go · TypeScript · AI Chatbot · Vertical Farming · Notification Services

- Architected an automated intelligent ecosystem for vertical farming optimization aimed at combating food insecurity.
- Integrated AI agents (GopherMind AI) and a conversational chatbot for automated monitoring and real-time agricultural data processing.
- Implemented a critical notification and alert system for automated crop resource control.
- Leading the technical paper writing under IEEE standards for international publication.

TeamFlow

2026

TypeScript · React · Git/Commit Management · Email Automation · Production

- Designed and built a platform for managing, tracking and optimizing collaborative workflows in development teams.
- Implemented a commit control module to centralize repository progress and audit code history from the UI.
- Built an automated email notification system to alert team members about task assignments and changes.

Axion Reality

2026

TypeScript · React · Artificial Intelligence · Augmented Reality (AR) · 3D Modeling

- Developed an immersive virtual environment management platform with AI for automated resource generation and a 3D model search engine.
- Implemented cross-platform Augmented Reality (AR) enabling users to visualize and interact with 3D models in real environments via QR code scanning.
- Optimized load performance and interactive graphics processing for a high-fidelity production interface.

Web Interface with Voice Control, Gestures and 3D Earth

2025

TypeScript · Firebase · MediaPipe · Three.js · Web Speech API · AI

- Built a real-time voice-controlled interface using the Web Speech API, with text-to-speech synthesis as the system response.
- Implemented gesture recognition with MediaPipe to navigate and control the page without touching the device.
- Built an interactive 3D Earth visualization with Three.js displaying geolocated points on the globe with immersive navigation.
- Integrated Firebase for authentication and persistence, with a voice-enabled AI conversational assistant as the navigation engine.

E-Commerce Copycaess

2026 – Present

PHP · MySQL · Composer · User roles · Google OAuth · PDF Export

- Built the Copycaess S.A.S e-commerce platform with full checkout, Excel-based inventory management and Google OAuth login.
- Implemented automated order notification emails, PDF invoice export and differentiated user roles with security standards.

ZettaShop — Dropshipping Platform (Personal Project)

2025 – Present

PHP · MySQL · Composer · User roles · Google OAuth · PDF Export

- Built a dropshipping platform with checkout, inventory management, automated emails, PDF invoices and Google login.
- Architected external supplier integration and order automation with a clean UI and applied security standards.

Digital School Magazine

2025

PHP · MySQL · CMS Panel · Production

- Built a full CMS for academic article publishing with an admin panel, authentication and category organization.

School Voting System

2025

PHP · MySQL · Authentication · Role control

- Implemented a platform to manage school elections with vote receipts and real-time statistics, enforcing one vote per student.

RFID Attendance Control System

2024

Arduino Uno · C/C++ · RFID · RTC · SD Module

- Designed a device that identifies users via RFID cards, logs date/time with RTC and exports Excel-compatible records.

Mobile App — Remote Control for ESP32 Robots

2025

Java · Android Studio · Bluetooth · ESP32

- Built an Android app to pair and control multiple ESP32 robots via Bluetooth with real-time commands.

Soccer Robot with ESP32 (Competition)

2025

ESP32 · L298N Driver · PWM · 18650 Batteries

- Programmed a robot with PWM speed control and autonomous movement; awarded 1st place at the 2nd regional robotics competition.

WORK EXPERIENCE

Full-Stack Web Developer

2025 – Present

Copypcaess S.A.S — Neiva, Colombia

- Construí desde cero la plataforma de e-commerce en producción — logré pasar de operación manual a sistema automatizado implementando checkout completo, gestión de inventario, correos transaccionales de pedidos y exportación de facturas en PDF, usando PHP MVC, MySQL y Composer.
- Blindé el acceso de la plataforma implementando OAuth (Google Login), control de roles diferenciados y validación de sesiones bajo buenas prácticas OWASP — gestionando el ciclo de vida completo del código con Git (ramas, commits, revisiones) y entregando mejoras de forma iterativa en metodología Agile.

EDUCATION

Software Engineering

2026 – Present

Universidad Surcolombiana — Neiva, Colombia

- Presenter at InnovaSoft Fair · 3rd place Regional Hackathon · Speaker at AI conference — all during the first semester.

Technical Degree in Software Development

2024 – 2025

SENA — Neiva, Colombia

- Training in software architecture, relational databases, full-stack web development and application security.

Technical High School Diploma

2020 – 2025

I.E. Ricardo Borrero Álvarez — Neiva, Colombia

CERTIFICATIONS

Foundations of Cybersecurity <i>Google / Coursera</i>	Sep 2025
Maximize Productivity With AI Tools <i>Google / Coursera</i>	Sep 2025
Python in Practice <i>Python Week — DAXUS LATAM</i>	Apr 2026
English — Intermediate Level (B1/B2) <i>Instituto YES — Neiva, Colombia</i>	2023 – 2024

LANGUAGES

Spanish: Native **English:** Intermediate — B2 listening comprehension, B1 reading and writing